Hugo Marquerie 22/01/2026

Obs holomorfa conjugada antiholomorfa

Observación 1. Sea $f \in \mathcal{H}(\Omega)$, entonces $\overline{f}$ es antiholomorfa en Ω y, además, $\forall z \in \Omega :$

$$\begin{aligned}\frac{\partial \overline{f}}{\partial \overline{z}}(z) &= \frac{1}{2}\left(\frac{\partial \overline{f}}{\partial x}(z) + i\,\frac{\partial \overline{f}}{\partial y}(z)\right) = \frac{1}{2}\left(\frac{\partial u}{\partial x}(z) - i\,\frac{\partial v}{\partial x}(z) + i\,\frac{\partial u}{\partial y}(z) + \frac{\partial v}{\partial y}(z)\right)\\ &\overset{\overset{\text{C-R}}{\downarrow}}{=} \frac{1}{2}\left(\frac{\partial u}{\partial x}(z) - i\,\frac{\partial v}{\partial x}(z) + i\left(-\frac{\partial v}{\partial x}(z)\right) + \frac{\partial u}{\partial x}(z)\right) = \frac{1}{2}\left(2\,\frac{\partial u}{\partial x}(z) - 2i\,\frac{\partial v}{\partial x}(z)\right)\\ &= \frac{\partial u}{\partial x}(z) - i\,\frac{\partial v}{\partial x}(z) = \overline{\frac{\partial f}{\partial z}(z)} = \overline{f'(z)}.\end{aligned}$$